Statistics Before Programs: Block ℓ_p-Norms for Cyclic Join Bounds

Extended version: full proofs

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This note accompanies the ICDE 2027 submission and gives the proofs that the paper states in condensed form: the ℓ_p Composition Inequality (the validity of every LpH-CEG bound) and the Cyclic Improvement theorem. Notation follows the paper.

1 The ℓ_p Composition Inequality

Setup. A hyperpath $P = (S_0{=}\emptyset, F_1, S_1, \ldots, F_m, S_m)$ adds a block of relations F_j at step j, $S_j = S_{j-1} \cup F_j$, and S_m is the set of all relations. Write $A_j = \mathrm{attr}(F_j)$, $V_j = \mathrm{attr}(S_j)$ ($V_0 = \emptyset$, $V_m = V$), the interface $I_j = A_j \cap V_{j-1}$ and $E_j = A_j \setminus I_j$. Then

(F1) $E_j = V_j \setminus V_{j-1}$, so $\{E_j\}_j$ partitions V;

(F2) $I_j \subseteq V_{j-1}$.

Weights $w_{j,p} \geq 0$ ($p \in \mathcal{P} \subseteq [1, \infty]$) satisfy the attribute-wise cover constraint of the paper: for every attribute $a \in V$, bound at step $j(a)$ (i.e. $a \in E_{j(a)}$),

$$w_{j(a)} + \sum_{j' :\, a \in I_{j'}} s_{j'} \;\geq\; 1, \qquad w_j := \sum_p w_{j,p}, \quad s_j := \sum_p w_{j,p}/p \;\; (1/\infty = 0). \tag{C}$$

A *polymatroid* is $h : 2^V \to \mathbb{R}$ with $h(\emptyset) = 0$, monotone and submodular; $h(X \mid Y) := h(X \cup Y) - h(Y)$.

Lemma 1 (Conditioning)**.** *For a polymatroid h and $Y \subseteq Z$: $h(X \mid Y) \geq h(X \mid Z)$.*

Proof. Submodularity on $X \cup Y$ and Z gives $h(X \cup Y) + h(Z) \geq h(X \cup Y \cup Z) + h((X \cup Y) \cap Z)$, and $h((X \cup Y) \cap Z) \geq h(Y)$ by monotonicity since $Y \subseteq (X \cup Y) \cap Z$. Rearrange. □

Lemma 2 (Fractional subadditivity)**.** *Let f be monotone submodular on subsets of a finite set with $f(\emptyset) = 0$, and let $\lambda_i \geq 0$ satisfy $\sum_{i:\, a \in T_i} \lambda_i \geq 1$ for every $a \in S$. Then $\sum_i \lambda_i f(T_i) \geq f(S)$.*

Proof. Order $S = (a_1, \ldots, a_k)$ and let $S_{<t} = \{a_1, \ldots, a_{t-1}\}$. By monotonicity, the chain rule and Lemma 1, $f(T_i) \geq f(T_i \cap S) = \sum_{a_t \in T_i} f(a_t \mid T_i \cap S_{<t}) \geq \sum_{a_t \in T_i} f(a_t \mid S_{<t})$. Weighting by λ_i and exchanging sums, $\sum_i \lambda_i f(T_i) \geq \sum_t f(a_t \mid S_{<t}) \sum_{i:\, a_t \in T_i} \lambda_i \geq \sum_t f(a_t \mid S_{<t}) = f(S)$, using $f(a_t \mid S_{<t}) \geq 0$. □

Lemma 3 (The obligation)**.** *For every polymatroid h and every w satisfying (C),*

$$\sum_j \Big[s_j \, h(I_j) + w_j \, h(E_j \mid I_j) \Big] \;\geq\; h(V).$$

Proof. Let $g_j = h(E_j \mid V_{j-1})$, $\delta_j = \max(0, 1 - w_j) \in [0, 1]$ and $f_j(S) = h(S \mid V_{j-1})$ for $S \subseteq E_j$; f_j is monotone, submodular and $f_j(\emptyset) = 0$.

(i) For each j', the chain rule along the partition (F1) and Lemma 1 give $h(I_{j'}) = \sum_j h(I_{j'} \cap E_j \mid I_{j'} \cap V_{j-1}) \geq \sum_j f_j(I_{j'} \cap E_j)$. Hence $\sum_{j'} s_{j'} h(I_{j'}) \geq \sum_j \sum_{j'} s_{j'} f_j(I_{j'} \cap E_j)$.

(ii) Fix j. If $\delta_j > 0$ then by (C) every $a \in E_j$ has $\sum_{j':a \in I_{j'}} s_{j'} \geq 1 - w_j = \delta_j$, so the sets $I_{j'} \cap E_j$ with weights $s_{j'}/\delta_j$ fractionally cover E_j, and Lemma 2 gives $\sum_{j'} s_{j'} f_j(I_{j'} \cap E_j) \geq \delta_j \, g_j$. If $\delta_j = 0$ the same inequality holds because the left side is non-negative.

(iii) By (F2) and Lemma 1, $h(E_j \mid I_j) \geq g_j \geq 0$, and $w_j \geq 1 - \delta_j$, so $w_j h(E_j \mid I_j) \geq (1 - \delta_j) g_j$.

Adding (i)–(iii): the left side is at least $\sum_j [\delta_j + (1 - \delta_j)] g_j = \sum_j h(E_j \mid V_{j-1}) = h(V)$ by the chain rule and (F1). □

Theorem 1 (ℓ_p Composition Inequality)**.** *Let D be a database, $\mathcal{Q}(D) \neq \emptyset$, and let $\mu_p(F_j \mid I_j)$ be the ℓ_p-norm of the degree sequence of E_j-values per I_j-value in a relation $R_j \supseteq \pi_{A_j} \mathcal{Q}(D)$—for instance the block join $\bowtie_{R \in F_j} R$ computed exactly on D. If w satisfies (C), then*

$$|\mathcal{Q}(D)| \;\leq\; \prod_{j=1}^{m} \prod_{p \in \mathcal{P}} \mu_p(F_j \mid I_j)^{\,w_{j,p}}.$$

Proof. Let h be the entropy of the uniform distribution on $\mathcal{Q}(D)$; it is a polymatroid and $h(V) = \log_2 |\mathcal{Q}(D)|$. The marginal on A_j is supported on $\pi_{A_j} \mathcal{Q}(D) \subseteq R_j$, so the ℓ_p degree inequality of Abo Khamis et al. (PODS 2024) gives, for $p \in [1, \infty]$, $\frac{1}{p} h(I_j) + h(E_j \mid I_j) \leq \log_2 \mu_p(F_j \mid I_j)$. Multiply by $w_{j,p} \geq 0$, sum, and apply Lemma 3. □

Scope. The proof uses only monotonicity and submodularity, so it holds over the whole polymatroid cone and for every hyperpath, block size and $p \geq 1$. It requires (C) exactly: an LP solver may return weights violating (C) by about 10^{-7}, which is removed by scaling w by $1/\min_a(\text{coverage of } a)$. It requires exact statistics of the database being bounded (or deterministic upper bounds on them, since the bound is monotone in each μ); a sampled or sketched statistic, such as the Hutchinson estimate of the common-neighbour ℓ_2 in the paper, is outside the guarantee. When some block join is empty the query is empty and the bound is 0.

Why the elementary argument does not suffice. If $w_j \geq 1$ for every step, Lemma 3 follows from (iii) alone. The LP's optima on cyclic queries have $w_j < 1$ (e.g. 1/3 on the closing step of a cycle); the deficit δ_j is paid for by later interfaces, which is exactly step (ii). The cyclic improvement and the need for Lemma 2 have the same source.

Independent checks. An LP over the polymatroid cone decides Lemma 3 at fixed weights; enumerating every vertex of the feasible weight region decides it for all weights. Over 6,197 hyperpath structures on eleven shapes with $|V| \leq 6$ and block sizes 1–3 the minimum slack is exactly 0; an adversarial search over random abstract hyperpath schemes with $|V| \leq 5$ found no violation; and over 207,000 databases the bound never fell below the exact count (`scripts/shannon_check_wide.py`, `scripts/validity_search.py`, `docs/step5_lemma.md`).

2 The Cyclic Improvement theorem

Order the cycle C_k as $R_1(a_1, a_2), \ldots, R_k(a_k, a_1)$ and extend in this order with single-relation blocks. Let $N_j = |R_j|$, M_j the maximum degree and $\mu_p(R_j)$ the ℓ_p-norm of the degree sequence of R_j from a_j to a_{j+1}.

Structure. Step 1 has interface $\emptyset$ and extension $\{a_1, a_2\}$: its degree sequence is the single value N_1. A middle step $2 \le j \le k-1$ has interface $\{a_j\}$, extension $\{a_{j+1}\}$ and $\mu_\infty = M_j$. The closing step k has interface $\{a_k, a_1\}$ and empty extension; its conditional degrees are all 1, so $\mu_\infty = 1$ and $\mu_1 = N_k$. With $p = \infty$ and unit weights this gives $B_{\max} = N_1 \prod_{j=2}^{k-1} M_j$.

A feasible point for every common $p \ge 1$. Put weight $\frac{p}{p+1}$ with exponent p on every middle step, $\frac{p}{p+1}$ on step 1 and $\frac{1}{p+1}$ on the closing step with exponent 1. The cover constraints read: a_1: $\frac{p}{p+1} + \frac{1}{p+1} = 1$; a_2: $\frac{p}{p+1} + \frac{p/(p+1)}{p} = 1$; a_i for $3 \le i \le k-1$: $\frac{p}{p+1} + \frac{1}{p+1} = 1$; a_k: $\frac{p}{p+1} + \frac{1}{p+1} = 1$. All are tight, so the LP optimum satisfies

$$B_{\mathrm{LP}} \le N_1^{\frac{p}{p+1}} N_k^{\frac{1}{p+1}} \prod_{j=2}^{k-1} \mu_p(R_j)^{\frac{p}{p+1}} \quad \text{and} \quad \frac{B_{\max}}{B_{\mathrm{LP}}} \ge \Big(\frac{N_1}{N_k}\Big)^{\frac{1}{p+1}} \prod_{j=2}^{k-1} \frac{M_j}{\mu_p(R_j)^{p/(p+1)}}.$$

The exponent must be common (or non-increasing). The constraint of the attribute between middle edges j and $j{+}1$ is $\frac{p_j}{p_j+1} + \frac{1}{p_{j+1}+1} \ge 1$, i.e. $p_{j+1} \le p_j$. Choosing each edge's best exponent independently is therefore not feasible in general (e.g. $p_j = 1$, $p_{j+1} = \infty$ covers that attribute only to $\frac{1}{2}$); the LP may still mix exponents within a step.

Growth. At $p = 2$ a middle factor $M_j/\mu_2(R_j)^{2/3}$ exceeds 1 exactly when $\mu_2(R_j) < M_j^{3/2}$, i.e. for skewed relations, so under fixed skew the ratio grows geometrically in the number $k-2$ of middle edges. As $p \to \infty$ every factor tends to 1, and LpH-CEG returns $\min(B_{\max}, B_{\mathrm{LP}})$, so the realized ratio is always at least 1. The boundary factor $(N_1/N_k)^{1/(p+1)}$ does not depend on k; it is 1 for a uniform cycle.

Acyclic queries. On a star the cover constraints force unit weight on every step and the LP returns the max-degree bound. On a path the LP can use an interface at $p = 1$ in place of an extension (on a 4-path it returns weights $(1, 0, 1)$), so acyclicity alone does not force unit weights; what redundant coverage on cycles adds is the slack exploited above.

3 Complexity

Collecting $\mu_p(F \mid I)$ for a single relation is one scan, $O(N)$; for a two-relation block it materializes the pair join, except for the interfaces computable from single-relation degree tables (all but the common-neighbour interface of a binary block; Section VI-C of the paper). The cover LP has $m|\mathcal{P}|$ variables and $|V|$ constraints; it solves in milliseconds for the queries in the paper. The flow LP over a $bs \le 2$ catalog is polynomial in the number of constraints and solves in 0.03–4.4 s at $k = 3$–16 over $4k$ orders.